

Some specialised wireless applications

The applications of wireless technologies that we have seen so far are limited to a specific region or to the transfer of small amounts of data without the use of wires. However, the applications that we are about to see now transcend all such petty boundaries. From a wireless system that spans the whole globe to a system that can actually power our appliances and devices without the use of wires, the forthcoming technologies truly boggle the mind.

First we have the global positioning system (GPS); this is a navigational system that has transformed the very meaning of the term. It can be used anywhere in the world and can help you pinpoint your location on the globe at any given point of time. The other is a fledgling technology which involves wireless power transfer. Although still in its initial stages, the technology has been quite encouraging so far and the day is not very far off when wires will become a thing of the past and we will be able to transfer energy by air alone. These are therefore technologies that have the capability to change the very face of civilisation as we know it.

8.1 Global positioning system

The global positioning system (GPS) is widely regarded as one of the most important inventions of the current era. It provides civilians with the capability to find out their exact location on the Earth from anywhere. It can also help in finding out distances covered in a specified time, rate of descent or ascent, and determining paths to specified locations.

The global positioning system traces its roots back to a navigation system called Transit. First successfully tested in 1960, it was used by the United States Navy. A very basic system, it involved just five satellites. Using these satellites, it could provide a satellite fix approximately once every hour. A major breakthrough in the history of GPS came in 1967, when the US navy developed the Timation satellite. The importance